

Quantum Mechanics II Exercise 6

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1 The Proca field: part II

The Proca theory describes a massive 4-vector field:

$$\mathcal{L} = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} + \frac{1}{2}m^2 B^\mu B_\mu \quad F_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu \quad (1)$$

With the conjugate momenta:

$$\Pi^\mu = \frac{\partial \mathcal{L}}{\partial \dot{B}_\mu} \rightarrow \Pi^0 = 0 \quad \Pi^i = \dot{B}_i - \partial_i B_0 \quad (2)$$

In this section we quantize the theory.

1.1 The Hamiltonian density

So, using a Legendre transformation, the Hamiltonian density is:

$$\mathcal{H} = \Pi^\mu \dot{B}_\mu - \mathcal{L} = \Pi^\mu \dot{B}_\mu + \frac{1}{4}F^{\mu\nu}F_{\mu\nu} - \frac{1}{2}m^2 B^\mu B_\mu \quad (3)$$

The first term follows:

$$\Pi^\mu \dot{B}_\mu = \Pi^i \dot{B}_i = \Pi^i (\Pi^i + \partial_i B_0) = \Pi^i \Pi^i + \Pi^i \partial_i B_0 \quad (4)$$

While, noticing that

$$\Pi^i = F^{i0} = -F_{i0} = -F^{0i} = F_{0i} \quad (5)$$

we get:

$$\frac{1}{4}F^{\mu\nu}F_{\mu\nu} = \frac{1}{4}F^{i0}F_{i0} + \frac{1}{4}F^{0i}F_{0i} + \frac{1}{4}F^{ij}F_{ij} = -\frac{1}{2}\Pi^i \Pi^i - \frac{1}{4}F_{ij}F_{ij} \quad (6)$$

And the last term:

$$\frac{1}{2}m^2 B^\mu B_\mu = \frac{1}{2}m^2 B_0^2 - \frac{1}{2}m^2 B_i B_i \quad (7)$$

Thus, we get:

$$\mathcal{H} = \Pi^i \Pi^i + \Pi^i \partial_i B_0 - \frac{1}{2}\Pi^i \Pi^i + \frac{1}{4}F_{ij}F_{ij} + \frac{1}{2}m^2 B_i B_i - \frac{1}{2}m^2 B_0^2 \quad (8)$$

Remembering that $\mathcal{H}$ fits in a spatial integral, we can integrate the second term by parts and ignore the total derivative term:

$$\Pi^i \partial_i B_0 = \partial_i (B_0 \Pi^i) - B_0 \partial_i \Pi^i \rightarrow -B_0 \partial_i \Pi^i \quad (9)$$

Giving:

$$\mathcal{H} = \frac{1}{2}\Pi^i \Pi^i + \frac{1}{4}F_{ij}F_{ij} + \frac{1}{2}m^2 B_i B_i - \frac{1}{2}m^2 B_0^2 - B_0 \partial_i \Pi^i \quad (10)$$

1.2 Equation of Motion for B_0

We can use E-L to find the EOM for B_0 , but this will require opening up the Lagrangian. Let us instead use Hamiltonian mechanics, since we got a comfortable Hamiltonian here. For the field B_0 , the conjugate momentum is zero, and so, using one of Hamilton's equations:

$$\dot{\Pi}^0 = \frac{\partial \mathcal{H}}{\partial B_0} \quad (11)$$

But since $\Pi^0 = 0$ we get the LHS of the equation is $\dot{\Pi}^0 = 0$. the RHS gives:

$$\frac{\partial \mathcal{H}}{\partial B_0} = -m^2 B_0 - \partial_i \Pi^i = 0 \quad (12)$$

This gives:

$$B_0 = -\frac{1}{m^2} \partial_i \Pi^i \quad (13)$$

Pluggin this back into $\mathcal{H}$:

$$\begin{aligned} \mathcal{H} &= \frac{1}{2} \Pi^i \Pi^i + \frac{1}{4} F_{ij} F_{ij} + \frac{1}{2} m^2 B_i B_i - \frac{1}{2} m^2 \left(\frac{1}{m^2} \partial_i \Pi^i \right)^2 + \frac{1}{m^2} \partial_j \Pi^j \partial_i \Pi_i \\ &= \frac{1}{2} \Pi^i \Pi^i + \frac{1}{4} F_{ij} F_{ij} + \frac{1}{2} m^2 B_i B_i + \frac{1}{2m^2} (\partial_i \Pi^i)^2 \end{aligned} \quad (14)$$

1.3 Quantizing the theory

Using the equal-time commutation relations:

$$\begin{aligned} [B_i(t, \vec{x}), \Pi_j(t, \vec{y})] &= i \delta_{ij}^3(\vec{x} - \vec{y}) \\ [B_i(t, \vec{x}), B_j(t, \vec{y})] &= [\Pi_i(t, \vec{x}), \Pi_j(t, \vec{y})] = 0 \end{aligned} \quad (15)$$

Remembering that the EOMs are:

$$\partial_\mu B^\mu = 0 \quad (\square + m^2) B^\mu = 0 \quad (16)$$

This is solved by the mode expansion:

$$B^\mu(x) = \sum_{\lambda=1}^3 \int \frac{d^3 p}{(2\pi)^3 \sqrt{2\omega_{\vec{p}}}} \left(\epsilon_\lambda^\mu(p) e^{-ipx} a_{\lambda, \vec{p}} + \epsilon_\lambda^\mu(p) e^{ipx} a_{\lambda, \vec{p}}^\dagger \right) \quad (17)$$

Where $\epsilon_\lambda^\mu(p)$ are the polarization vectors for $\lambda = 1, 2, 3$ satisfy:

- Transveraslity: $p_\mu \epsilon_\lambda^\mu(p) = 0$
- Orthonormality: $\epsilon_\lambda^\mu(p) \epsilon_{\lambda', \mu}(p) = -\delta_{\lambda\lambda'}$
- Completeness: $\sum_{\lambda=1}^3 \epsilon_\lambda^\mu(p) \epsilon_\lambda^\nu(p) = -\eta^{\mu\nu} + \frac{p^\mu p^\nu}{m^2}$

To see this works, we plug it into the EOMs:

$$\begin{aligned} \partial_\mu B^\mu &= \partial_\mu \sum_{\lambda=1}^3 \int \frac{d^3 p}{(2\pi)^3 \sqrt{2\omega_{\vec{p}}}} \left(\epsilon_\lambda^\mu(p) e^{-ipx} a_{\lambda, \vec{p}} + \epsilon_\lambda^\mu(p) e^{ipx} a_{\lambda, \vec{p}}^\dagger \right) = \\ &= \sum_{\lambda=1}^3 \int \frac{d^3 p}{(2\pi)^3 \sqrt{2\omega_{\vec{p}}}} \left(-ip_\mu \epsilon_\lambda^\mu(p) e^{-ipx} a_{\lambda, \vec{p}} + ip_\mu \epsilon_\lambda^\mu(p) e^{ipx} a_{\lambda, \vec{p}}^\dagger \right) = 0 \end{aligned} \quad (18)$$

Where in the last step we used the transversality of the polarization vectors. For the KG equation:

$$\begin{aligned} (\square + m^2) B^\mu &= (\partial_\nu \partial^\nu + m^2) \sum_{\lambda=1}^3 \int \frac{d^3 p}{(2\pi)^3 \sqrt{2\omega_{\vec{p}}}} \left(\epsilon_\lambda^\mu(p) e^{-ipx} a_{\lambda,\vec{p}} + \epsilon_\lambda^\mu(p) e^{ipx} a_{\lambda,\vec{p}}^\dagger \right) = \\ &= (-p_\nu p^\nu + m^2) \sum_{\lambda=1}^3 \int \frac{d^3 p}{(2\pi)^3 \sqrt{2\omega_{\vec{p}}}} \left(\epsilon_\lambda^\mu(p) e^{-ipx} a_{\lambda,\vec{p}} + \epsilon_\lambda^\mu(p) e^{ipx} a_{\lambda,\vec{p}}^\dagger \right) = 0 \end{aligned} \quad (19)$$

Where we used $m^2 = p_\nu p^\nu$. To prove the commutation relations, let us find the conjugate momenta:

$$\begin{aligned} \Pi^i &= \partial_0 B_i - \partial_i B_0 = \\ &= \sum_{\lambda=1}^3 \int \frac{d^3 p}{(2\pi)^3 \sqrt{2\omega_{\vec{p}}}} \left[ip^0 \left(-\epsilon_\lambda^i(p) e^{-ipx} a_{\lambda,\vec{p}} + \epsilon_\lambda^i(p) e^{ipx} a_{\lambda,\vec{p}}^\dagger \right) - ip^i \left(-\epsilon_\lambda^0(p) e^{-ipx} a_{\lambda,\vec{p}} + \epsilon_\lambda^0(p) e^{ipx} a_{\lambda,\vec{p}}^\dagger \right) \right] = \\ &= i \sum_{\lambda=1}^3 \int \frac{d^3 p}{(2\pi)^3 \sqrt{2\omega_{\vec{p}}}} \left[- (p^0 \epsilon_\lambda^i(p) - p^i \epsilon_\lambda^0(p)) e^{-ipx} a_{\lambda,\vec{p}} + (p^0 \epsilon_\lambda^i(p) - p^i \epsilon_\lambda^0(p)) e^{ipx} a_{\lambda,\vec{p}}^\dagger \right] = \\ &= i \sum_{\lambda=1}^3 \int \frac{d^3 p}{(2\pi)^3 \sqrt{2\omega_{\vec{p}}}} (p^0 \epsilon_\lambda^i(p) - p^i \epsilon_\lambda^0(p)) \left(-e^{-ipx} a_{\lambda,\vec{p}} + e^{ipx} a_{\lambda,\vec{p}}^\dagger \right) \end{aligned} \quad (20)$$

Using the commutation relations:

$$\left[a_{\lambda,\vec{p}}, a_{\lambda',\vec{k}}^\dagger \right] = \delta_{\lambda\lambda'} (2\pi)^3 \delta^3(\vec{p} - \vec{k}) \quad \left[a_{\lambda,\vec{p}}, a_{\lambda',\vec{k}} \right] = \left[a_{\lambda,\vec{p}}^\dagger, a_{\lambda',\vec{k}}^\dagger \right] = 0 \quad (21)$$

We get:

$$\begin{aligned} [B_i, B_j] &= \sum_{\lambda,\lambda'=1}^3 \int \frac{d^3 p d^3 k}{(2\pi)^6 2 \sqrt{\omega_{\vec{p}} \omega_{\vec{k}}}} \left(\epsilon_\lambda^i(p) \epsilon_{\lambda'}^j(\vec{k}) e^{-i(px-ky)} \left[a_{\lambda,\vec{p}}, a_{\lambda',\vec{k}}^\dagger \right] + \epsilon_\lambda^i(p) \epsilon_{\lambda'}^j(\vec{k}) e^{-i(ky-px)} \left[a_{\lambda,\vec{p}}^\dagger, a_{\lambda',\vec{k}} \right] \right) = \\ &= \sum_{\lambda=1}^3 \int \frac{d^3 p}{(2\pi)^3 2\omega_{\vec{p}}} \left(\epsilon_\lambda^i(p) \epsilon_\lambda^j(p) e^{-ip(x-y)} + \epsilon_\lambda^i(p) \epsilon_\lambda^j(p) e^{-ip(y-x)} \right) = 0 \end{aligned} \quad (22)$$

Where we concluded it's zero since the first and second terms are opposite. The same will happen for $[\Pi_i, \Pi_j]$. Let us define the shorthand notation $\alpha_\lambda^i(p) = p^0 \epsilon_\lambda^i(p) - p^i \epsilon_\lambda^0(p)$ to get:

$$\Pi^i = i \sum_{\lambda=1}^3 \int \frac{d^3 p}{(2\pi)^3 \sqrt{2\omega_{\vec{p}}}} \left(-\alpha_\lambda^i(p) e^{-ipx} a_{\lambda,\vec{p}} + \alpha_\lambda^i(p) e^{ipx} a_{\lambda,\vec{p}}^\dagger \right)$$

And so:

$$\begin{aligned} [\Pi^i, \Pi^j] &= - \sum_{\lambda,\lambda'=1}^3 \int \frac{d^3 p d^3 k}{(2\pi)^6 2 \sqrt{\omega_{\vec{p}} \omega_{\vec{k}}}} \left(\alpha_\lambda^i(p) \alpha_{\lambda'}^j(\vec{k}) e^{-i(px-ky)} \left[a_{\lambda,\vec{p}}, a_{\lambda',\vec{k}}^\dagger \right] + \alpha_\lambda^i(p) \alpha_{\lambda'}^j(\vec{k}) e^{-i(ky-px)} \left[a_{\lambda,\vec{p}}^\dagger, a_{\lambda',\vec{k}} \right] \right) = \\ &= \sum_{\lambda=1}^3 \int \frac{d^3 p}{(2\pi)^3 2\omega_{\vec{p}}} \left(\alpha_\lambda^i(p) \alpha_\lambda^j(p) e^{-ip(x-y)} + \alpha_\lambda^i(p) \alpha_\lambda^j(p) e^{-ip(y-x)} \right) = 0 \end{aligned}$$

And finally:

$$\begin{aligned} [B^i, \Pi^i] &= i \sum_{\lambda,\lambda'=1}^3 \int \frac{d^3 p d^3 k}{(2\pi)^6 2 \sqrt{\omega_{\vec{p}} \omega_{\vec{k}}}} \left(\epsilon_\lambda^i(p) \alpha_{\lambda'}^j(\vec{k}) e^{-i(px-ky)} \left[a_{\lambda,\vec{p}}, a_{\lambda',\vec{k}}^\dagger \right] - \epsilon_\lambda^i(p) \alpha_{\lambda'}^j(\vec{k}) e^{-i(ky-px)} \left[a_{\lambda,\vec{p}}^\dagger, a_{\lambda',\vec{k}} \right] \right) = \\ &= i \sum_{\lambda=1}^3 \int \frac{d^3 p}{(2\pi)^3 2\omega_{\vec{p}}} \left(\epsilon_\lambda^i(p) \alpha_\lambda^j(p) e^{-ip(x-y)} - \epsilon_\lambda^i(p) \alpha_\lambda^j(p) e^{-ip(y-x)} \right) = \\ &= i \sum_{\lambda=1}^3 \int \frac{d^3 p}{(2\pi)^3 2\omega_{\vec{p}}} \left(\epsilon_\lambda^i(p) \alpha_\lambda^j(p) e^{-ip(x-y)} + \epsilon_\lambda^i(p) \alpha_\lambda^j(p) e^{-ip(x-y)} \right) \end{aligned}$$

But, from the completeness relation of $\epsilon_\lambda^i(p)$ we get:

$$\begin{aligned}
\sum_{\lambda=1}^3 \epsilon_\lambda^i(p) \alpha_\lambda^j(p) &= \sum_{\lambda=1}^3 \epsilon_\lambda^i(p) \left[p^0 \epsilon_\lambda^j(p) - p^j \epsilon_\lambda^0(p) \right] = \\
&= p^0 \sum_{\lambda=1}^3 \epsilon_\lambda^i(p) \epsilon_\lambda^j(p) - p^j \sum_{\lambda=1}^3 \epsilon_\lambda^i(p) \epsilon_\lambda^0(p) = \\
&= p^0 \left[\eta^{ij} - \frac{p^i p^j}{m^2} \right] - p^j \left[\eta^{i0} - \frac{p^i p^0}{m^2} \right] = \\
&= p^0 \eta^{ij} - p^j \eta^{i0} = \omega_{\vec{p}} \delta^{ij}
\end{aligned}$$

Plugging that back into the integral gives:

$$[B^i, \Pi^j] = i \int \frac{d^3 p}{(2\pi)^3 2\omega_{\vec{p}}} 2\omega_{\vec{p}} \delta^{ij} e^{-ip(x-y)} = i \delta_{ij} \delta(\vec{x} - \vec{y})$$

1.4 Correlation function

Let us now derive the time-ordered 2-points correlation function:

$$D_F^{\mu\nu}(x-y) = \langle 0 | T B^\mu(x) B^\nu(y) | 0 \rangle$$

The time-ordering says that we assume once that $x^0 > y^0$ and order the fields accordingly, and then do the opposite, and add the results with Heaviside functions:

$$T B^\mu(x) B^\nu(y) = \Theta(x^0 - y^0) B^\mu(x) B^\nu(y) + \Theta(y^0 - x^0) B^\mu(y) B^\nu(x)$$

Now, let us evaluate each of the cases. In the $x^0 > y^0$ case:

$$\begin{aligned}
\langle 0 | T B^\mu(x) B^\nu(y) | 0 \rangle &= \langle 0 | B^\mu(x) B^\nu(y) | 0 \rangle = \sum_{\lambda, \lambda'=1}^3 \int \frac{d^3 p d^3 k}{(2\pi)^6 2 \sqrt{\omega_p \omega_k}} \epsilon_\lambda^\mu(p) \epsilon_{\lambda'}^\nu(k) e^{i(ky - px)} \langle 0 | a_{\lambda, p} a_{\lambda', k}^\dagger | 0 \rangle = \\
&= \sum_{\lambda, \lambda'=1}^3 \int \frac{d^3 p d^3 k}{(2\pi)^6 2 \sqrt{\omega_p \omega_k}} \epsilon_\lambda^\mu(p) \epsilon_{\lambda'}^\nu(k) e^{i(ky - px)} \langle 0 | [a_{\lambda, p}, a_{\lambda', k}^\dagger] | 0 \rangle = \\
&= \sum_{\lambda, \lambda'=1}^3 \int \frac{d^3 p d^3 k}{(2\pi)^6 2 \sqrt{\omega_p \omega_k}} \epsilon_\lambda^\mu(p) \epsilon_{\lambda'}^\nu(k) e^{i(ky - px)} \delta_{\lambda\lambda'} (2\pi)^3 \delta(\vec{p} - \vec{k}) = \\
&= \sum_{\lambda=1}^3 \int \frac{d^3 k}{(2\pi)^3 2\omega_k} \epsilon_\lambda^\mu(k) \epsilon_\lambda^\nu(k) e^{ik(y-x)} = \\
&= \int \frac{d^3 k}{(2\pi)^3 2\omega_k} e^{-ik(x-y)} \left(-\eta^{\mu\nu} + \frac{k^\mu k^\nu}{m^2} \right)
\end{aligned}$$

And of course, for the case $y^0 > x^0$ we get the same just with $x \leftrightarrow y$. Let us now show that the four-integral form

$$D_F^{\mu\nu}(x-y) = \lim_{\epsilon \rightarrow 0^+} \int \frac{d^4 k}{(2\pi)^4} \frac{-ie^{-ik(x-y)}}{k^2 - m^2 + i\epsilon} \left(\eta^{\mu\nu} - \frac{k^\mu k^\nu}{m^2} \right)$$

Is equivalent. To show that, we perform the k^0 integral using the residue theorem. First let us separate the spatial integral:

$$\lim_{\epsilon \rightarrow 0^+} \int \frac{d^4 k}{(2\pi)^4} \frac{-ie^{-ik(x-y)}}{k^2 - m^2 + i\epsilon} \left(\eta^{\mu\nu} - \frac{k^\mu k^\nu}{m^2} \right) = \int \frac{d^3 k}{(2\pi)^3} e^{i\vec{k} \cdot (\vec{x} - \vec{y})} \lim_{\epsilon \rightarrow 0^+} \int \frac{-ie^{-ik^0(x^0 - y^0)}}{k^2 - m^2 + i\epsilon} \left(\eta^{\mu\nu} - \frac{k^\mu k^\nu}{m^2} \right) \quad (23)$$

And let us now look at the k^0 integral. The denominator becomes:

$$k^2 - m^2 + i\varepsilon = (k^0)^2 - \vec{k}^2 - m^2 + i\varepsilon = (k^0)^2 - \omega_k^2 + i\varepsilon \rightarrow (k^0 - (\omega_k - i\varepsilon)) (k^0 + (\omega_k - i\varepsilon)) \quad (24)$$

here in the last step we renamed ε and added a ε^2 term that vanishes in the limit. This means the integrand has two poles $k^0 = \pm(\omega_k - i\varepsilon)$. Since one of them is above the real axis and one of them is below, when we want to integrate k^0 in $(-\infty, \infty)$, we need to take the semi circle contour below or above the axis, depending on which of them doesn't diverge. Looking at the exponent $e^{-ik^0(x^0 - y^0)}$, we get that for $x^0 > y^0$, the integral diverges when $\text{Im}(k^0) > 0$ and vice versa. So, we take the below contour if $x^0 > y^0$ and the above contour if $x^0 < y^0$. The below contour contains the pole $k^0 = \omega_k - i\varepsilon$ while the above contour contains the pole $k^0 = -\omega_k + i\varepsilon$. Let us calculate the residue in each pole: